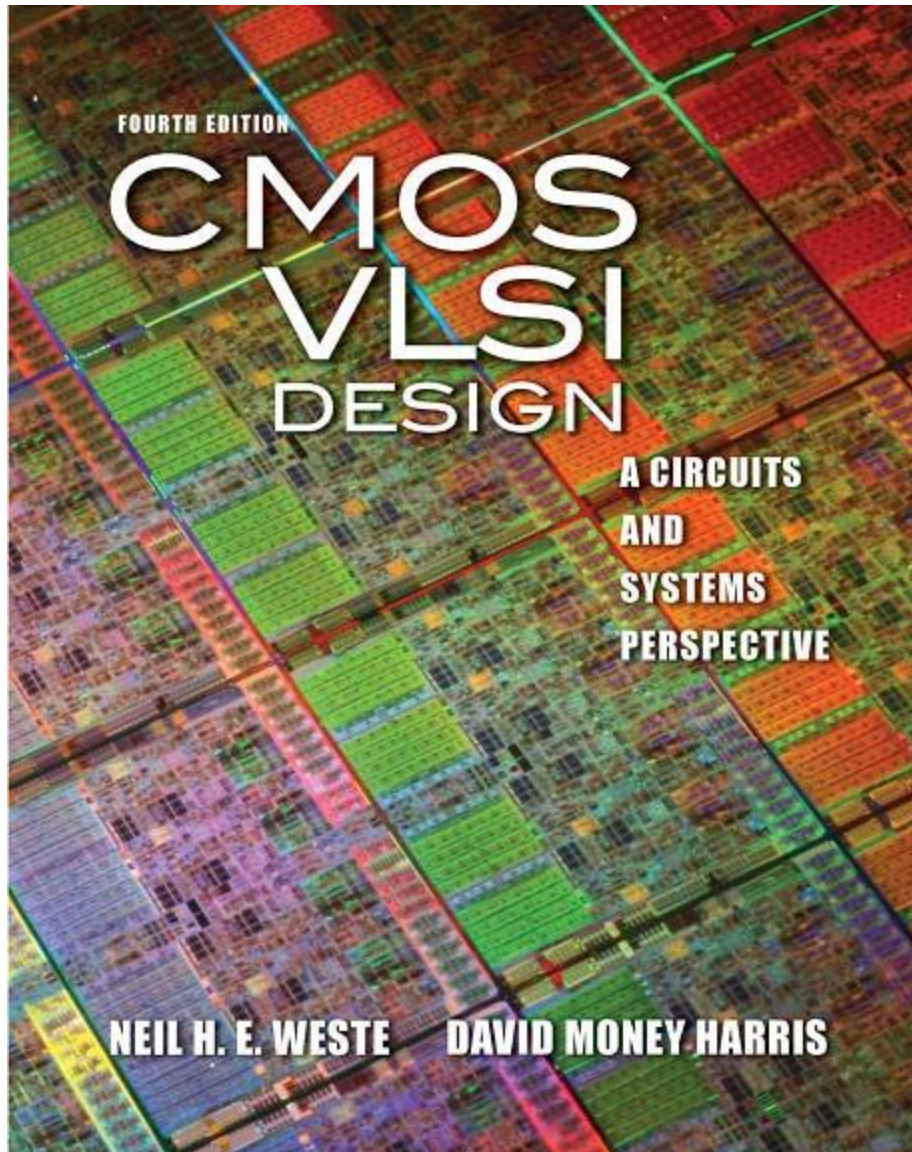


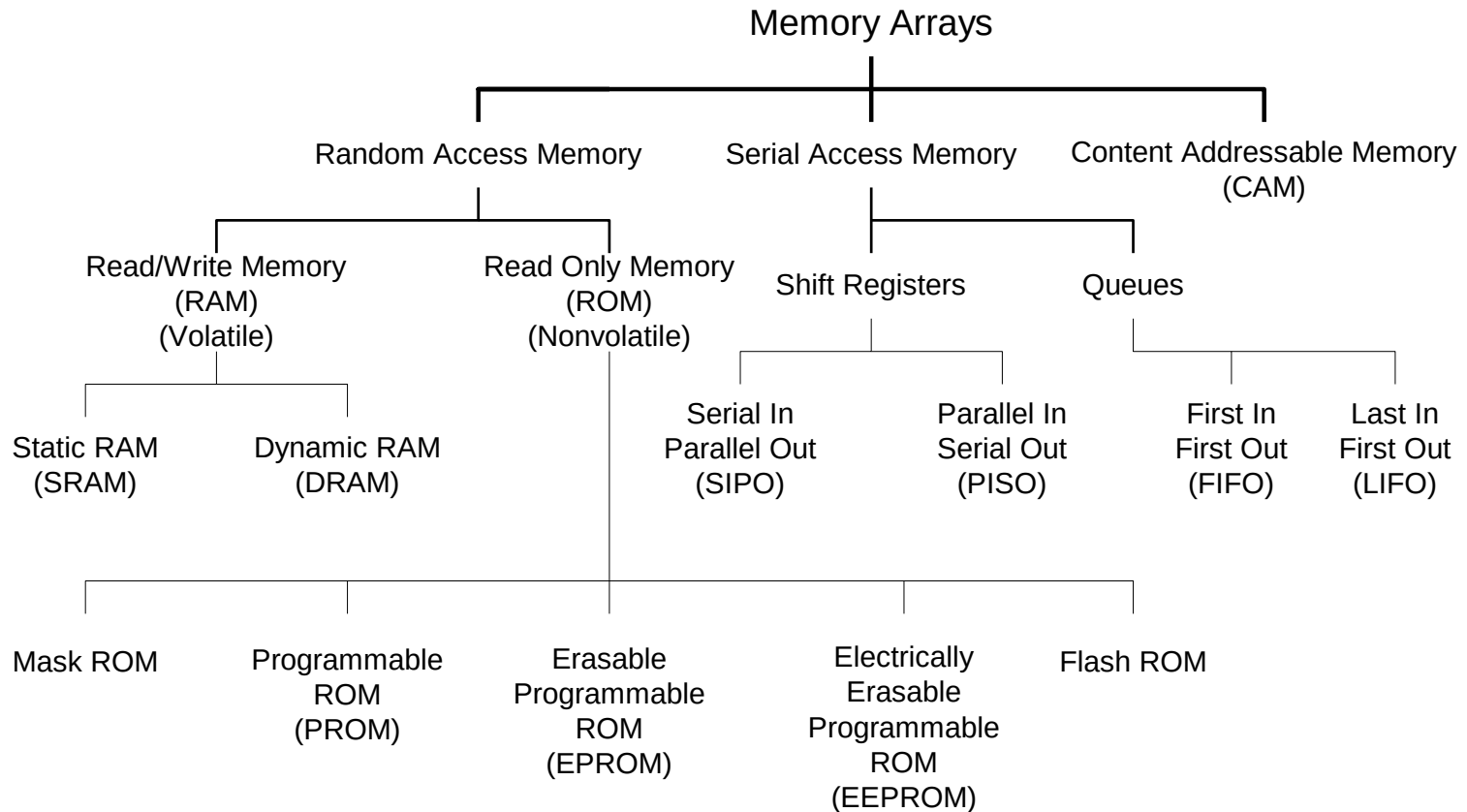


Array Subsystems

Lecture # 3

- ☐ Read-Only Memories
- ☐ Mask-programmed ROMs
- ☐ ROM Array
- ☐ PROMs and EPROMs
- ☐ Flash Programming
- ☐ NAND Flash

Memory Arrays



Non-volatile Memories

- ❑ The contents of a mask ROM are hardwired during fabrication and cannot be changed.
- ❑ A programmable ROM (PROM) can be programmed once after fabrication by blowing on-chip fuses with a special high programming voltage.
- ❑ An erasable programmable ROM (EPROM) is programmed by storing charge on a floating gate. It can be erased by exposure to ultraviolet (UV) light for several minutes to knock the charge off the gate. Then the EPROM can be reprogrammed.
- ❑ Electrically erasable programmable ROMs (EEPROMs) are similar, but can be erased in microseconds with on-chip circuitry.
- ❑ Flash memories are a variant of EEPROM that erases entire blocks rather than individual bits. Sharing the erase circuitry across larger blocks reduces the area per bit. Because of their good density and easy in system reprogrammability, Flash memories have replaced other nonvolatile memories in most modern CMOS systems.

Read-Only Memories

- ❑ Read-Only Memories are nonvolatile
 - Retain their contents when power is removed
- ❑ Mask-programmed ROMs use one transistor per bit
 - Presence or absence determines 1 or 0

ROM Example

❑ 4-word x 6-bit ROM

- Represented with dot diagram
- Dots indicate 1's in ROM

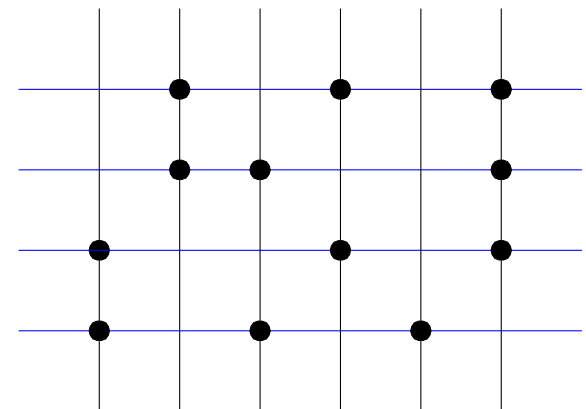
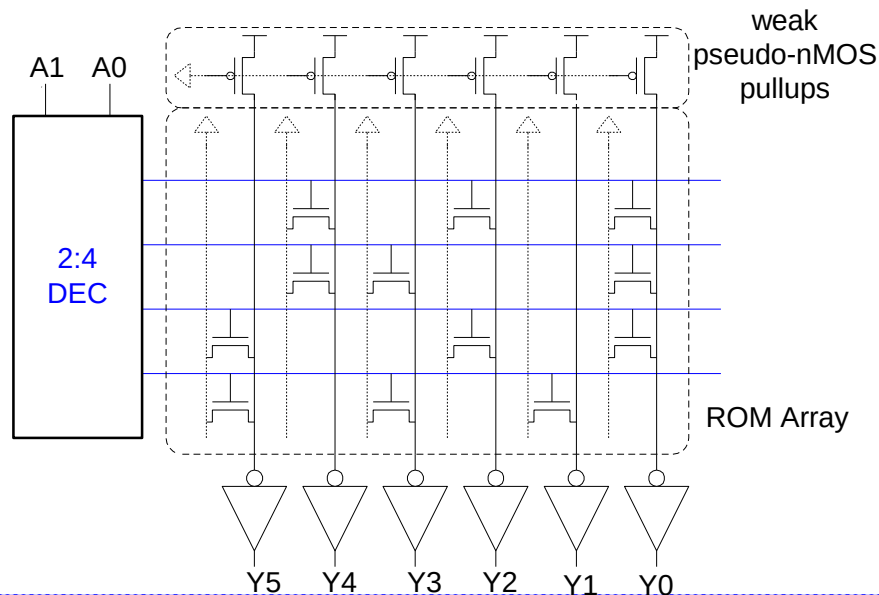
❑ Looks like 6 4-input pseudo-nMOS NORs

Word 0: **010101**

Word 1: **011001**

Word 2: **100101**

Word 3: **101010**



ROM Example

- ❑ 4-word x 6-bit ROM
- ❑ Looks like 6 4-input pseudo-nMOS NANDs

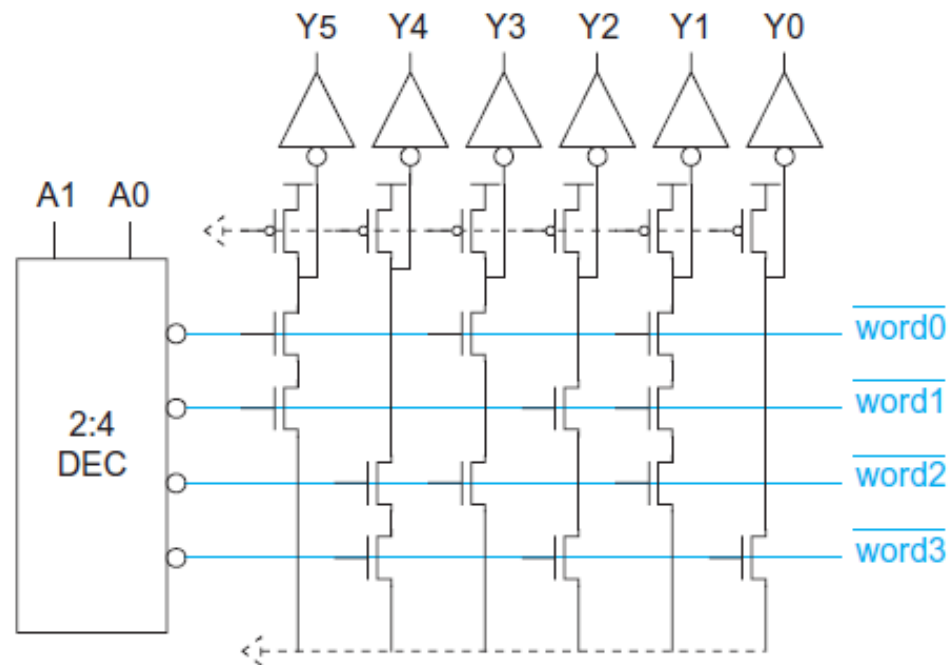
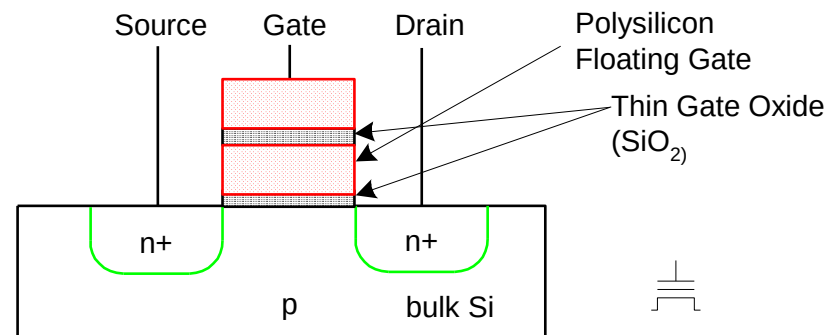


FIGURE 12.58 Pseudo-nMOS NAND ROM

PROMs and EPROMs

- ❑ Programmable ROMs
 - Build array with transistors at every site
 - Burn out fuses to disable unwanted transistors
- ❑ Erasable Programmable ROMs
 - Use floating gate to turn off unwanted transistors
 - EPROM, EEPROM, Flash



Programming to 0

- ❑ High voltage to the Control Gate (CG) and grounding the substrate
- ❑ Induces negative voltage onto the Floating Gate (FG)
- ❑ Raises the V_t of the transistor that it is always OFF
- ❑ Because V_t has been raised so it is Positive V_t
- ❑ Represents Logic 0

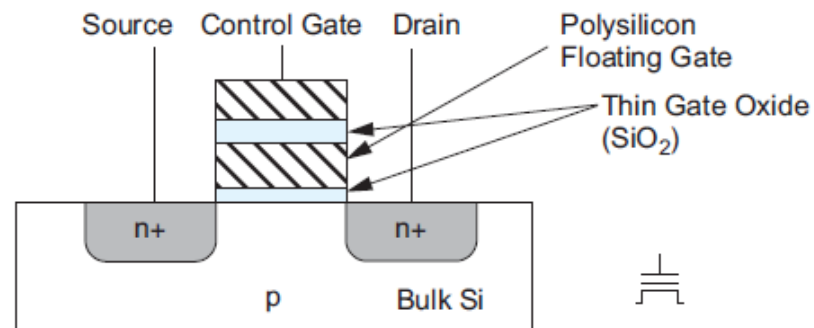


FIGURE 12.57 Cross-section of floating gate nMOS transistor

Erasing to 1

- ❑ High voltage to the substrate and grounding the Control Gate (CG)
- ❑ Removing negative voltage onto the Floating Gate (FG)
- ❑ Reduces the V_t of the transistor that it is always ON
- ❑ Because V_t has been reduced so it is Negative V_t
- ❑ Represents Logic 1

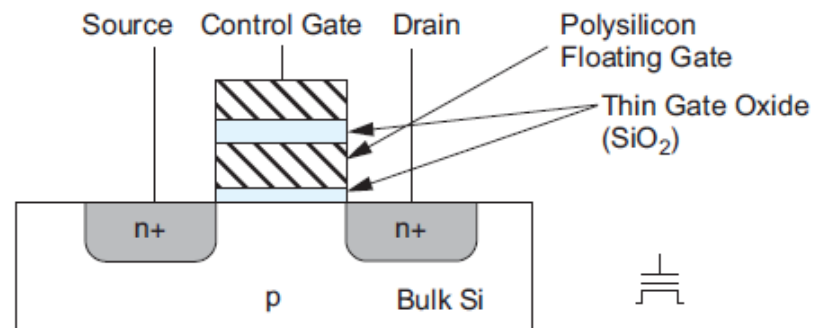
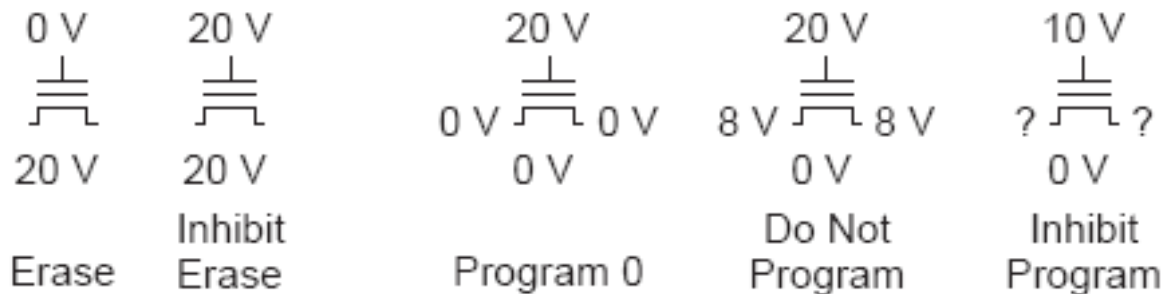


FIGURE 12.57 Cross-section of floating gate nMOS transistor

Flash Programming

- ❑ Charge on floating gate determines V_t
- ❑ Logic 1: negative V_t
- ❑ Logic 0: positive V_t
- ❑ Cells erased to 1 by applying a high body voltage so that electrons tunnel off floating gate into substrate
- ❑ Programmed to 0 by applying high gate voltage

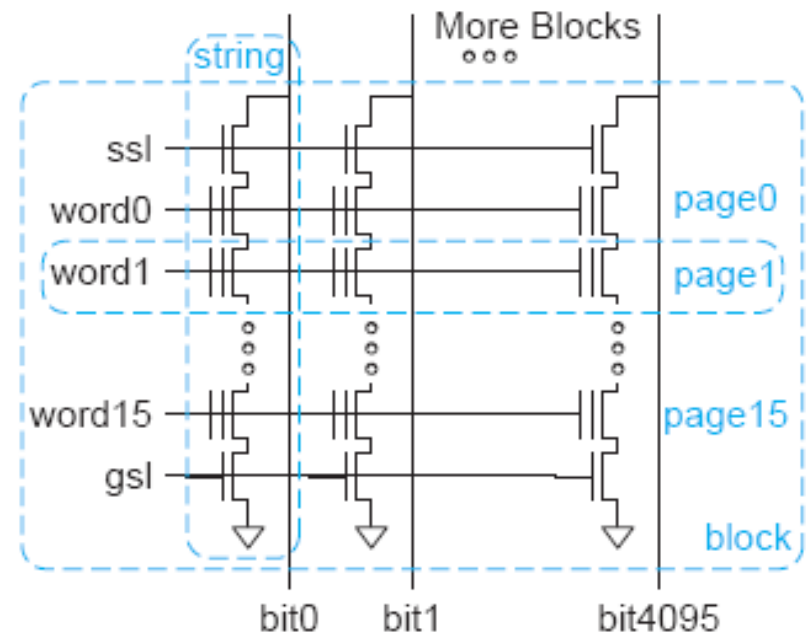


NAND Flash

- ❑ High density, low cost / bit
 - Programmed one page at a time
 - Erased one block at a time

- ❑ Example:

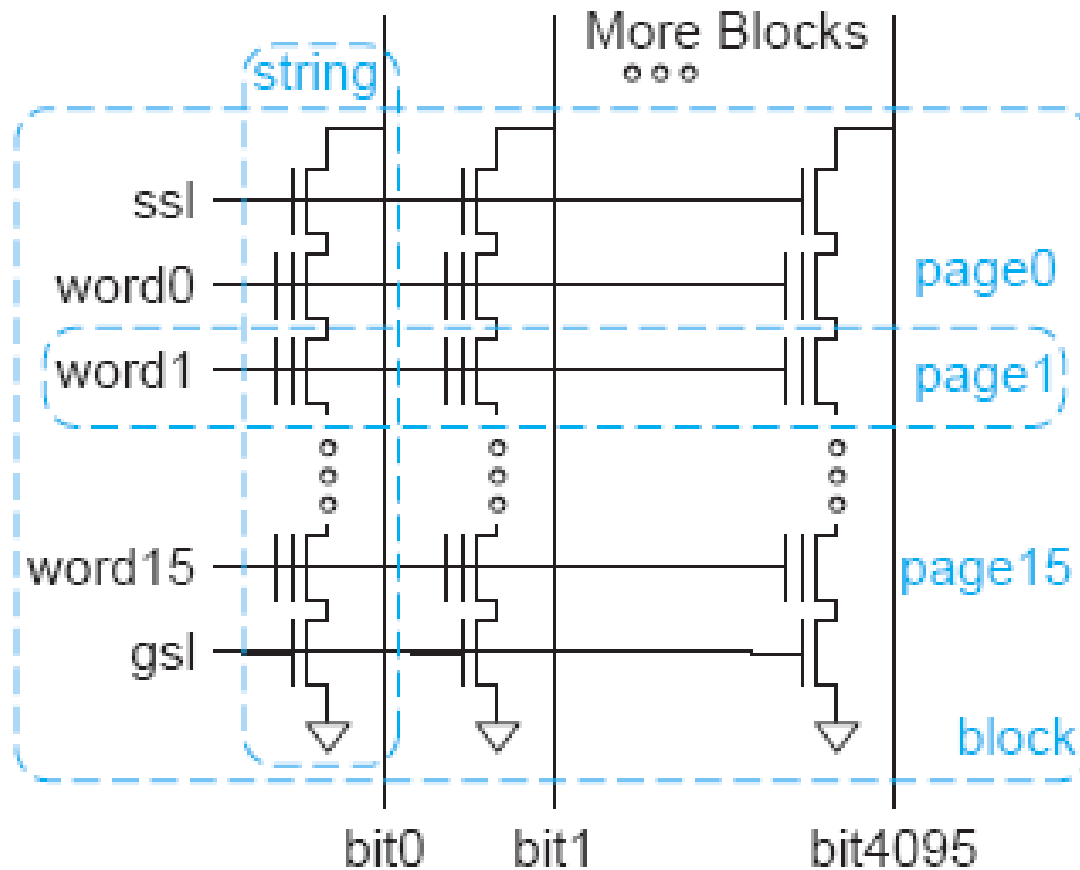
- 8 KB (64 Kb) block
- 16 pages / 8 KB block
- 512 B (4Kb) 4096-bit pages
- 16 Cells/string
- Many blocks / memory



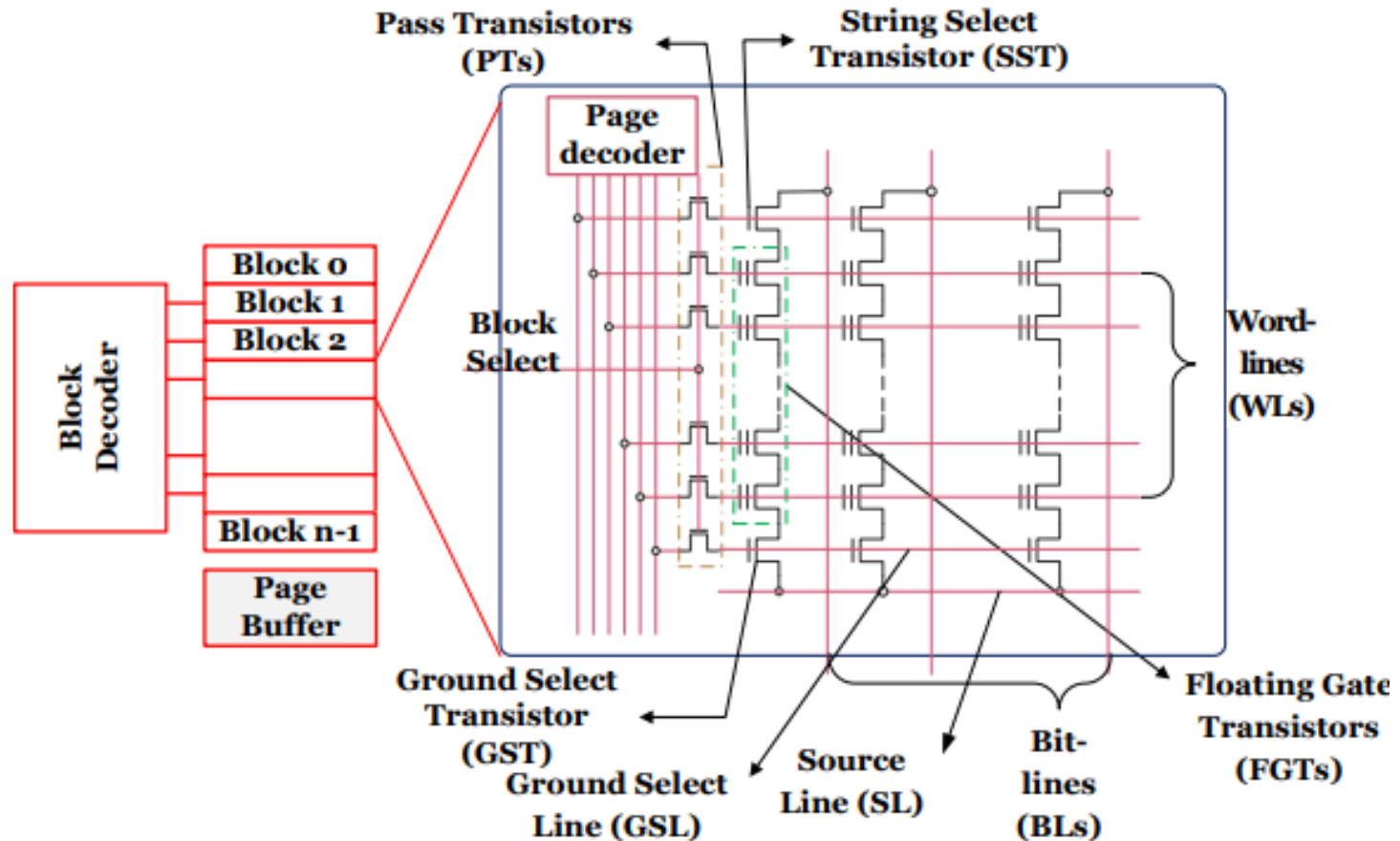
Reading Operation

- ❑ A page is read in a similar fashion to a conventional NAND ROM.
- ❑ The bitlines are precharged.
- ❑ ssl and gsl are both set to 3.3 V to activate the selected block.
- ❑ The active-low wordline for the selected page is set to 0 V and the wordlines for all the other pages in the block are set to 4.5 V, which is much higher than V_t .
- ❑ Thus, all the transistors in the stack are ON except possibly the one corresponding to the selected page.
- ❑ If the cell being read has a negative V_t , it turns ON too and the bitline discharges.
- ❑ If the cell being read has a positive V_t , it remains OFF and the bitline does not switch.

Reading Operation



NAND Flash Array



64 Gb NAND Flash

- ❑ 64K cells / page
- ❑ 4 bits / cell (multiple V_t)
- ❑ 64 cells / string
 - 256 pages / block
- ❑ 2K blocks / plane
- ❑ 2 planes



[Trinh09]

Home Work

- Design a 16 KB NAND Flash memory block, where each string contains 32 cells. How many number of pages are required to design this memory and what will be the size of each page.